Rotational Dynamics

Physics, Class 12, Chapter 1 · Maharashtra Board (HSC) · Complete Notes · Weightage 5 marks (7 with option)

1. Introduction: Rotation vs Revolution

Axis of rotation: the fixed line about which every particle of the body moves in a circle.
Rotation: axis passes through (or within) the body — fan blades, top, Earth spinning on its axis.
Revolution: body moves around an external axis — Earth around the Sun.
Every particle of a rotating rigid body has the same angular displacement, angular velocity and angular acceleration, but different linear speeds (depends on its distance r from the axis).

2. Characteristics of Circular Motion

Why circular motion is accelerated: velocity is a vector; its direction changes at every instant along the circle, so velocity changes ⇒ acceleration exists, directed to the centre.
Centripetal (radial) acceleration:

$$\boxed{a_c = \frac{v^2}{r} = r\omega^2 = v\omega = \frac{4\pi^2 r}{T^2}}$$

Centripetal force: not a new force. It is the *requirement* of circular motion, supplied by a real force:

- stone on a string → tension; vehicle turning → friction; satellite → gravity; electron around nucleus → electrostatic force.

$$\boxed{F_c = \frac{mv^2}{r} = mr\omega^2}$$

Uniform circular motion (UCM): constant speed, constant angular speed; only direction changes. Periodic in space.
Period T (time per revolution) and **frequency** n (revolutions per second): $n = 1/T, \quad \omega = 2\pi n = 2\pi/T$.
Non-uniform circular motion: speed also changes ⇒ two perpendicular components of acceleration: centripetal $a_c = r\omega^2$ and tangential $a_t = r\alpha$. Net $a = \sqrt{a_c^2 + a_t^2}$.

3. Applications of Uniform Circular Motion

3.1 Vertical circular motion (non-uniform) Speed and tension both change with position (gravity slows the body going up, speeds it coming down). For a body of mass m on a string of length r:

Top: $T_{\text{top}} + mg = \dfrac{mv_t^2}{r}$ Bottom: $T_{\text{bot}} - mg = \dfrac{mv_b^2}{r}$

Critical (minimum) speed at the top (string just taut, $T_{\text{top}} = 0$):

$$\boxed{v_{\text{top}} = \sqrt{gr}}$$

Minimum speed at the bottom to complete the circle (energy conservation):

$$\boxed{v_{\text{bot}} = \sqrt{5gr}}$$

Useful results: v at any point from $v_b^2 = v_t^2 + 4gr$; tension difference $T_{\text{bot}} - T_{\text{top}} = 6mg$ (direct MCQ result).
String can only pull, not push ⇒ if $v_{\text{top}} < \sqrt{gr}$ the string slackens and the circular path breaks. A rigid rod can push, so its condition at the top is only $v_t \geq 0$.

3.2 Vehicle on a horizontal circular road Centripetal force is supplied entirely by friction: $\dfrac{mv^2}{r} \leq \mu_s mg$

$$\boxed{v_{\max} = \sqrt{\mu_s rg}}$$

3.3 Vehicle on a banked road Road is inclined by angle θ towards the centre. The normal reaction N tilts; its horizontal component supplies the centripetal force, so friction is not needed at the designed speed.
Banking relation (no friction):

$$\boxed{\tan\theta = \frac{v^2}{rg}}$$

Independent of the vehicle's mass — same designed speed for cycle and truck.
With friction, maximum safe speed:

$$\boxed{v_{\max} = \sqrt{\frac{rg(\mu_s + \tan\theta)}{1 - \mu_s \tan\theta)}}}$$

Above the designed speed the vehicle tends to skid *up* the slope, so friction acts *down* the slope and raises the safe limit.
Well of death: the wall's normal reaction provides centripetal force; friction balances the weight. Rider must maintain speed. **Diagrams to practise (draw in exam)** (1) Banked road: draw the inclined road, weight mg down, normal N perpendicular to the surface; resolve N into $N\cos\theta$ (balances mg) and $N\sin\theta$ (towards centre). (2) Vertical circle: mark tension and mg at the top (both downward) and at the bottom (opposite).

4. Kinematics of Rotational Motion

Angular displacement θ (rad): angle swept at the centre. 1 revolution $= 2\pi$ rad $= 360°$; 1 rad $\approx 57.3°$ (arc length $=$ radius).
Angular velocity $\omega = d\theta/dt$ (rad/s); **angular acceleration** $\alpha = d\omega/dt$ (rad/s^2).
RPM to rad/s: $\omega = \dfrac{2\pi N}{60}$.
Linear–angular bridge relations:

$$\boxed{v = r\omega \qquad a_t = r\alpha \qquad a_c = r\omega^2 = \frac{v^2}{r}}$$

Same ω at every point of a rigid body, but points farther from the axis move faster (fan tip fastest).
Kinematic equations (mirrors of linear motion):

$$\boxed{\omega = \omega_0 + \alpha t \qquad \theta = \omega_0 t + \tfrac{1}{2}\alpha t^2 \qquad \omega^2 = \omega_0^2 + 2\alpha\theta}$$

5. Torque (Moment of Force)

The turning effect of a force about an axis. Depends on: magnitude of force, perpendicular distance from the axis (lever arm), and the angle of application.

$$\boxed{\vec{\tau} = \vec{r} \times \vec{F} \qquad |\tau| = rF\sin\theta = F \times (\text{lever arm})}$$

Maximum when force $\perp$ lever arm; zero if the line of force passes through the axis. SI unit: N m (not joule). Vector along the axis; anticlockwise usually taken positive.
Couple: two equal, opposite, parallel, non-collinear forces. Net force $= 0$ (no translation), net torque $= F\times$ perpendicular separation (arm of couple) — pure rotation. Same torque about any axis. Examples: steering wheel, tap handle, screwdriver. **Work done by a torque:**

$$\boxed{W = \tau\theta \qquad P = \tau\omega}$$

Torque in terms of moment of inertia (board derivation) For a particle: $\tau = rF = r(ma_t) = r\,m(r\alpha) = mr^2\alpha$. Summing over all particles of the rigid body (same α):

$$\boxed{\tau = I\alpha \quad \text{(rotational form of Newton's second law)}}$$

6. Moment of Inertia and Radius of Gyration

Moment of inertia I**:** the rotational analogue of mass — the body's resistance to change in its rotational state.

$$\boxed{I = \sum_i m_i r_i^2 = \int r^2\,dm \qquad [\text{kg m}^2]}$$

Depends on mass *and* its distribution about the chosen axis; the same body has different I about different axes. Mass farther from the axis contributes much more (r^2).

Physical significance: in every linear equation, replacing m by I (and v, a, F, p by ω, α, τ, L) gives the rotational equation. I plays in rotation exactly the role mass plays in translation.

Radius of gyration K**:** the distance from the axis at which the whole mass could be concentrated to give the same I:

$$\boxed{I = MK^2}$$

Theorem of perpendicular axes (plane lamina only) For axes x, y in the plane and z perpendicular through their intersection:

$$\boxed{I_z = I_x + I_y}$$

Theorem of parallel axes (any body) I_c about the axis through the centre of mass; about any parallel axis at distance h:

$$\boxed{I = I_c + Mh^2}$$

I is minimum about the axis through the centre of mass.

Moments of inertia of standard bodies (memorise)

Body	Axis	I	K^2/R^2
Ring (radius R)	through centre, $\perp$ plane	MR^2	1
Disc / solid cylinder	through centre, $\perp$ plane	$\frac{1}{2}MR^2$	$1/2$
Hollow sphere	diameter	$\frac{2}{3}MR^2$	$2/3$
Solid sphere	diameter	$\frac{2}{5}MR^2$	$2/5$
Uniform rod (length L)	through centre, $\perp$ rod	$\frac{ML^2}{12}$	—
Uniform rod	through end, $\perp$ rod	$\frac{ML^2}{3}$	—

Rod about end follows from parallel axes: $\frac{ML^2}{12} + M\left(\frac{L}{2}\right)^2 = \frac{ML^2}{3}$. Board asks derivations for the ring and the rod.

7. Angular Momentum and Its Conservation

Angular momentum (moment of linear momentum): for a particle $\vec{L} = \vec{r} \times \vec{p}$, $L = mvr$; for a rigid body:

$$\boxed{L = I\omega \qquad [\text{kg m}^2/\text{s}]}$$

Torque as rate of change:

$$\boxed{\tau = \frac{dL}{dt} \;\; (\text{constant } I \Rightarrow \tau = I\alpha)}$$

Conservation of angular momentum: if net *external torque* is zero, $L = I\omega =$ constant ($I_1\omega_1 = I_2\omega_2$). External forces may exist; external torque must not.

- Ice skater / diver / ballet: pulling limbs in $\Rightarrow$ I falls $\Rightarrow$ ω rises.
- Planets: nearer the Sun (r smaller) $\Rightarrow$ faster (basis of Kepler's 2nd law).
- Helicopter tail rotor counters the reaction spin of the main rotor.
- Collapsing star $\rightarrow$ fast-spinning pulsar.

8. Rotational Work, Energy and Power

Rotational kinetic energy:

$$\boxed{KE_{\text{rot}} = \tfrac{1}{2}I\omega^2}$$

Work–energy theorem: net work of torques = change in rotational KE.
Flywheel: large-I wheel that stores energy as $\frac{1}{2}I\omega^2$ and returns it smoothly (engines, toys).
Linear–rotational analogy (master table):

Linear	Rotational
Displacement s	θ
Velocity v / acceleration a	ω / α
Mass m	Moment of inertia I
Force F	Torque τ
Momentum $p = mv$	Angular momentum $L = I\omega$
$F = ma$	$\tau = I\alpha$
$W = Fs, \;\; P = Fv$	$W = \tau\theta, \;\; P = \tau\omega$
$KE = \frac{1}{2}mv^2$	$KE = \frac{1}{2}I\omega^2$

9. Rolling Motion

Rolling = translation of the centre of mass + rotation about it.
Pure rolling (no slipping): the contact point is instantaneously at rest;

$$\boxed{v = r\omega}$$

Total kinetic energy:

$$\boxed{KE = \tfrac{1}{2}Mv^2 + \tfrac{1}{2}I\omega^2 = \tfrac{1}{2}Mv^2\left(1 + \frac{K^2}{R^2}\right)}$$

Rolling down an incline (height h, angle θ):

$$\boxed{v = \sqrt{\frac{2gh}{1 + K^2/R^2}} \qquad a = \frac{g\sin\theta}{1 + K^2/R^2}}$$

Race order (down an incline): smaller K^2/R^2 wins (less energy locked in spin):

$$\boxed{\text{solid sphere } (0.4) > \text{disc } (0.5) > \text{hollow sphere } (0.67) > \text{ring } (1)}$$

Mass and radius cancel — all solid spheres finish together.
Role of friction: static friction is *necessary* to start/maintain rolling but does *no work* in pure rolling (contact point at rest), so mechanical energy is conserved. On a frictionless incline the body only slides with $a = g\sin\theta$ — faster than any roller.

Board Exam Pointers

- Weightage: 5 marks (7 with option).
- Derivations to master: critical velocity of vertical circle; banking angle $\tan\theta = v^2/rg$; $\tau = I\alpha$; KE of rolling body / rolling on incline; parallel axis theorem; MI of ring and rod.
- Direct results: $T_{\text{bot}} - T_{\text{top}} = 6mg$; $v_{\max}$ on flat road $= \sqrt{\mu rg}$; race order.
- Numericals: RPM $\rightarrow$ rad/s conversions; kinematic equations; $I_1\omega_1 = I_2\omega_2$; $I = MK^2$ with parallel axes.

Appendix: Key Derivations (Board Favourites)

A. Critical speed at the top of a vertical circle At the top, weight and tension both point to the centre: $T + mg = \dfrac{mv^2}{r}$.

A string can only pull $\Rightarrow T \geq 0$. Minimum speed at $T = 0$: $mg = \dfrac{mv_{\text{top}}^2}{r} \Rightarrow \boxed{v_{\text{top}} = \sqrt{gr}}$.

For the bottom, energy conservation over height $2r$: $\tfrac{1}{2}mv_b^2 = \tfrac{1}{2}mv_t^2 + mg(2r)$; with $v_t^2 = gr$: $\boxed{v_b = \sqrt{5gr}}$.

B. Banking of roads On a road banked at angle θ, resolve the normal reaction N: vertical component balances weight, $N\cos\theta = mg$; horizontal component supplies centripetal force, $N\sin\theta = \dfrac{mv^2}{r}$. Dividing: $\boxed{\tan\theta = \dfrac{v^2}{rg}}$ (mass cancels).

C. Moment of inertia of a uniform ring (central axis) Every element dm of the ring is at the same distance R from the axis: $I = \displaystyle\int r^2\, dm = R^2 \int dm = \boxed{MR^2}$.

D. Moment of inertia of a uniform rod about its centre Linear density $\lambda = M/L$. Element at distance x from centre: $dI = x^2\, dm = \lambda x^2 dx$. Integrate from $-L/2$ to $+L/2$: $I = \lambda \displaystyle\int_{-L/2}^{L/2} x^2 dx = \lambda\left[\frac{x^3}{3}\right]_{-L/2}^{L/2} = \frac{M}{L}\cdot\frac{L^3}{12} = \boxed{\frac{ML^2}{12}}$.

E. Rolling down an incline (energy method) Starts from rest at height h: loss in PE = gain in total KE. $Mgh = \tfrac{1}{2}Mv^2 + \tfrac{1}{2}I\omega^2 = \tfrac{1}{2}Mv^2 + \tfrac{1}{2}MK^2\dfrac{v^2}{R^2} = \tfrac{1}{2}Mv^2\left(1 + \dfrac{K^2}{R^2}\right) \Rightarrow \boxed{v = \sqrt{\dfrac{2gh}{1 + K^2/R^2}}}$ and $a = \dfrac{g\sin\theta}{1 + K^2/R^2}$. Smaller $K^2/R^2 \Rightarrow$ larger v and a (race order).